

3.8.3 Limitations of the evaluation protocol

Results characterise *dispatch fidelity* and *mode-contract compliance* under snapshot grounding. They do not constitute a live Kaggle [DOM](#) study or a human expert rubric. Agentic scores therefore reflect planning and batching behaviour within the two-call cap rather than full autonomous notebook completion. A future release could add a ReAct-style observe–act loop (Yao et al., [2023](#)) with per-cell feedback; that capability is explicitly out of scope for the present version.

3.9 Interaction Mode Specifications

Each mode is enforced at three layers: the extension user-interface dropdown, host-side mode resolution in `streaming.py`, and prompt/tool-filter rules in `prompt_engineering.py`. [Table 3.2](#) states the behavioural contract evaluated in [Chapter 4](#).

Table 3.2: Interaction mode contracts.

Mode	Primary output	Tool policy	Notebook side effects
Ask	Natural-language explanation	No <code>tool_calls</code>	None
Code	Placement + python block	Prefetch reads only; no browser writes	None (user copies code)
Agentic	Tool batch + brief summary	Two-phase: query then implement	Dispatched via extension

3.9.1 Ask mode

Ask mode loads `prompts/ask.txt` and packs notebook context without attaching browser tool schemas. The model is instructed to explain cells, diagnose errors, and describe workflows using snapshot evidence (Rule et al., [2018](#); McNutt & DeLine, [2023](#)). Streaming delivers incremental prose to the chat panel. This mode aligns with read-only tutoring scenarios in educational contexts (Finnie-Ansley et al., [2022](#)). [Figure 3.7](#) illustrates the Ask mode advisory flow.

3.9.2 Code mode

Code mode loads `prompts/code.txt` and supports optional placement prefetch. Tools include `notebook_recommend_placement` and `notebook_find_symbol`. Replies are structured as